Simpson's Paradox



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The Setup: Choosing a Surgeon

- Imagine you need surgery and compare two doctors' track records:
  - **Dr. Jones:** 86% overall success rate
  - **Dr. Smith:** 79% overall success rate
- The numbers seem to make the choice for you

> ***Question:*** *Is picking the doctor with the higher success rate always the rational choice?*

Watch: Simpson's Paradox Explained

- Simpson's Paradox: Why More Data Can Trick You
  - Source of the surgeon example and worked math in this deck
- How Statistics Can Be Misleading — TED-Ed, Mark Liddell (2016)
  - 4 min; adds a smoking study and Florida death penalty cases
  - Ends on the point most explainers skip: the overall number is sometimes the right one

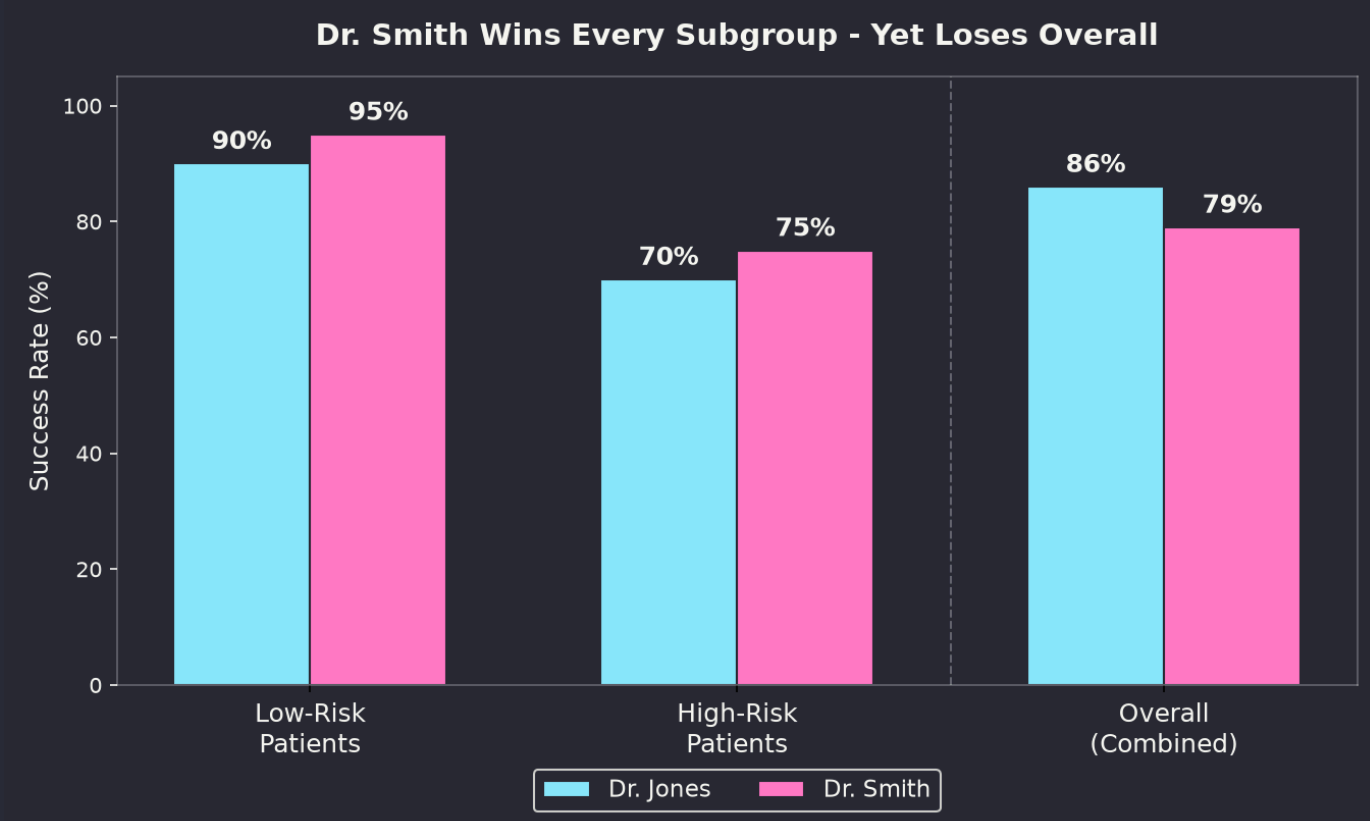
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Digging Deeper: Split by Patient Risk

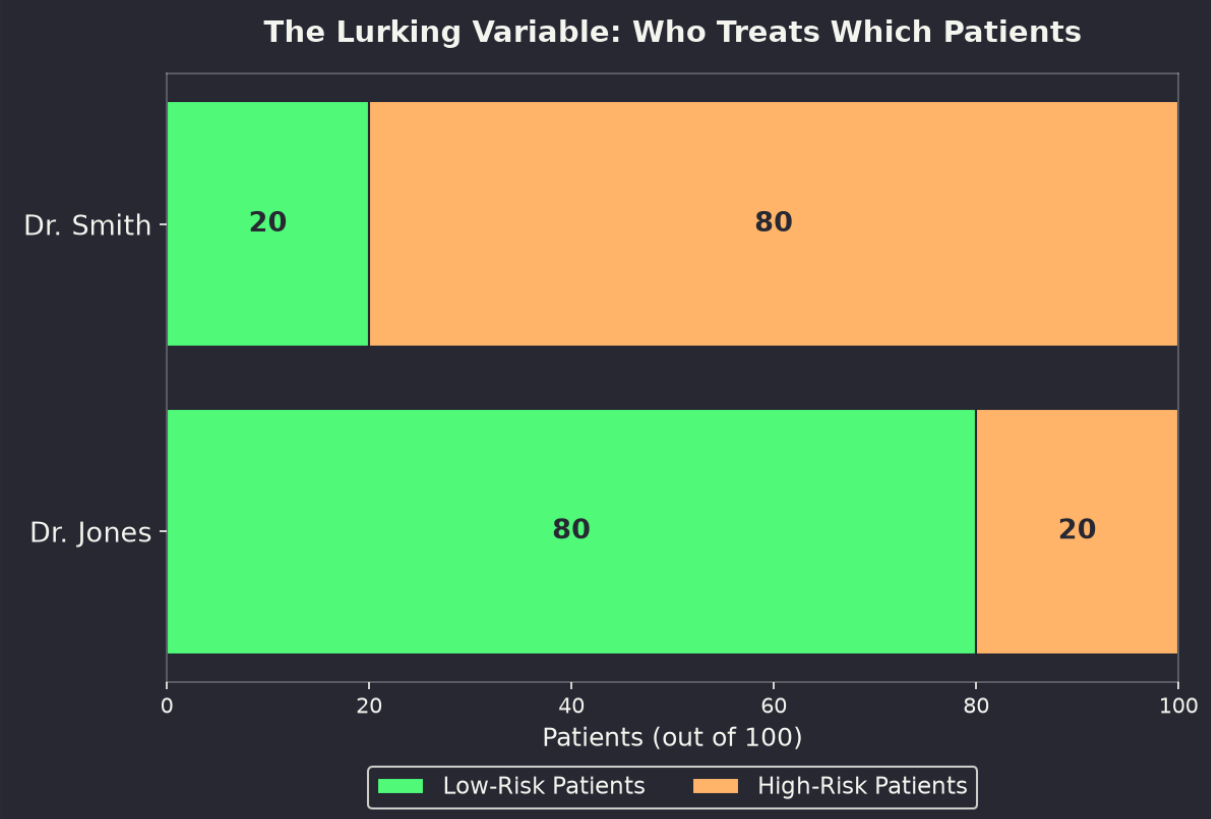
- Ask the hospital for outcomes broken down by patient type
- **High-risk patients:**
  - Dr. Smith: **75%** success
  - Dr. Jones: **70%** success
- **Low-risk patients:**
  - Dr. Smith: **95%** success
  - Dr. Jones: **90%** success
- Dr. Smith wins **both** subgroups — yet loses overall

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The Numbers, Visualized



How Is This Possible? The Caseload Mix



Doing the Math: Weighted Averages

Dr. Smith (20 low-risk, 80 high-risk)

- Low-risk: 20 × 95% = **19** survivors
- High-risk: 80 × 75% = **60** survivors
- Total: 79 / 100 = **79%**

Dr. Jones (80 low-risk, 20 high-risk)

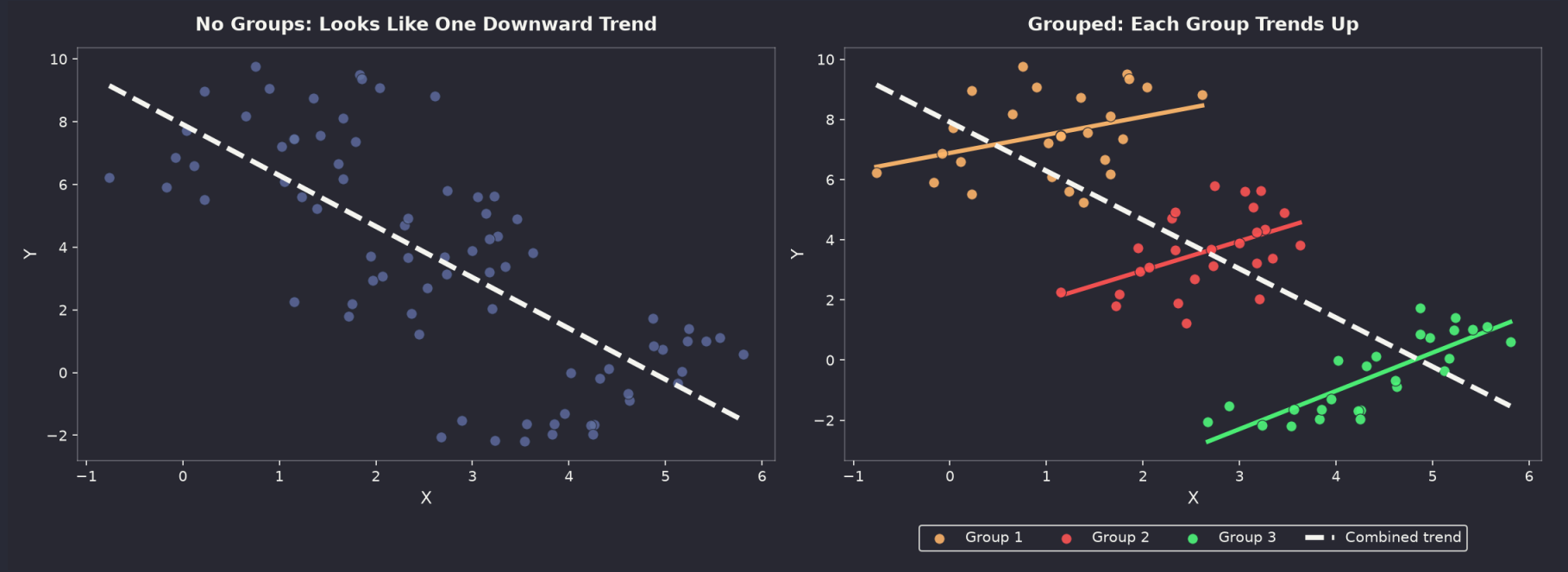
- Low-risk: 80 × 90% = **72** survivors
- High-risk: 20 × 70% = **14** survivors
- Total: 86 / 100 = **86%**

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Naming the Phenomenon

- **Simpson's Paradox:**
  - a trend appears in several groups of data
    - but disappears or reverses when the groups are combined
- Variables can be **positively correlated** within every subgroup
  - yet **negatively correlated** in aggregate
- The overall number told you to pick Jones
- The subgroup breakdown says Smith is the better choice,
  - **for either kind of patient**

Trend Reversal, Visualized



- **Left:** no group labels — looks like one clean downward trend
- **Right:** same data, colored by group — each group actually trends **upward**
- No single point moved — only whether the grouping was visible

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The Lurking Variable

- A **lurking (confounding) variable** is a hidden factor
  - that changes what the data appears to say
- In this example:
  - it is the **proportion of high-risk vs. low-risk patients** each surgeon treats
  - It's correlated with:
    - The **outcome** (high-risk patients survive less often, regardless of surgeon)
    - The **group** being compared (Smith sees far more high-risk cases)
  - Ignore it, and the aggregate comparison is comparing apples to oranges

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Real-World Case: Hospital Rankings

- Major referral centers and trauma hospitals
  - often look **worse** on raw survival-rate rankings
- Reason:
  - they take the sickest, most complex patients that smaller hospitals decline to treat
- Once outcomes are **risk-adjusted** for patient severity,
  - top referral centers often move up substantially
- Raw league tables can punish the hospitals doing the hardest, most valuable work

> *Source: Krumholz et al. (2006), risk-standardized mortality methodology — see References*

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Real-World Case: Kidney Stone Treatments (1986)

Success rate	Small stones	Large stones	Overall
Open surgery (older)	**93%** (81/87)	**73%** (192/263)	78% (273/350)
PCNL (newer, less invasive)	87% (234/270)	69% (55/80)	**83%** (289/350)

- The newer treatment looks **better overall** — but is **worse for both** stone sizes
- Cause: doctors gave the newer treatment mostly **small, easier** stones
- This was an **observational** comparison, not a randomized trial

> *Source: Charig et al. (1986), British Medical Journal — see References*

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Real-World Case: UC Berkeley Admissions (1973)

- Berkeley's graduate admissions showed an aggregate gap:
  - **44%** of male applicants admitted vs. **35%** of female applicants
- This looked like evidence of bias against women applicants
- Broken down **by department**,
  - most departments showed **no bias** — some even favored women slightly
- Explanation:
  - women applied in greater numbers to more competitive departments
    - with **lower overall admit rates**

> *Source: Bickel, Hammel & O'Connell (1975), Science — see References*

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Real-World Cases: Age as the Lurking Variable

Smoking and survival (UK)

- 20-year follow-up of 1,314 women in Whickham
- Overall: **24%** of smokers died vs **31%** of non-smokers
- The non-smokers were **older** on average
- Within most age groups, smokers died at the **higher** rate

COVID-19 case fatality (2020)

- Italy's overall case fatality rate was **higher** than China's
- Yet Italy's rate was **lower in every age group**
- Italy's confirmed cases were much **older**
- Same mechanism: a different mix, not a different risk

“ *Sources: Appleton, French & Vanderpump (1996); von Kügelgen, Gresele & Schölkopf (2021) — see References* ”

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Aggregates Can Deceive

- An overall rate is a **weighted average** of subgroup rates, weighted by group size
- A group with a **large, easy** subset can post a high overall number without being the best performer
- A group with a **large, hard** subset can post a low overall number despite being the best performer
- This isn't rare — it shows up in medicine, admissions, sports, sales, and A/B tests
- **Golden Rule:** an aggregate statistic is a starting question, not a final answer

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Look for Subgroups — Carefully

- Before trusting an aggregate comparison, ask:
  - What natural subgroups exist in this data?
  - Are they distributed **unevenly** across the things being compared?
  - Does splitting by subgroup change — or reverse — the conclusion?
- But splitting has its own risks:
  - Slice the data enough ways and **some split will reverse almost any result**
  - Small subgroups give **noisy** rates
- Choose splits for a reason **before** looking at the results

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Split or Combine? Ask What Causes What

- **Split** when the third variable affects **both** the group and the outcome
  - Patient risk → which surgeon *and* survival
  - Stone size → which treatment *and* success
- **Combine** when the third variable is **caused by** the thing being compared
  - A drug lowers blood pressure, and lower blood pressure aids recovery
  - Splitting by blood pressure hides part of the drug's real effect
- The same numbers can call for either answer
  - the data alone can't tell you which — **how the data was produced** does

> *Source: Pearl (2014), Understanding Simpson's Paradox — see References*

Before You Present a Comparison

1. List plausible lurking variables — and **how** each relates to the groups and the outcome
2. Compute both aggregate and subgroup rates; check whether the subgroup **weights** differ
3. Investigate any reversal — don't dismiss it as a data error
4. Decide split vs combine from **what causes what**, not from which answer you prefer
5. Report risk-adjusted or stratified metrics alongside raw ones
6. Never rank people, teams, or hospitals on raw aggregate rates alone

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Key Takeaway

- The aggregate number is never the whole story
- Always ask what's hiding underneath the average:
  - a lurking variable,
  - an uneven mix,
  - a hidden group

> *Sometimes the better choice looks worse on paper*

References

Topic	Source
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